

PROFESSIONAL SUMMARY

PhD candidate in theoretical particle physics with **three** peer-reviewed publications in Physical Review D (**50+** citations) and with invited talks at Brookhaven National Lab and CERN. Specializing in large-scale numerical computation, symbolic manipulation, and multi-parameter optimization applied to high-dimensional scientific datasets to explore physics beyond the standard model. Extensive experience developing analytical frameworks in Mathematica and high-performance numerical workflows in Python and C++. Interested in scientific machine learning, physics-informed neural networks, and quantum information science.

EDUCATION

- **Stony Brook University** Aug 2020 – Present
Ph.D. in Physics (High Energy Theory and Phenomenology)
- **Stony Brook University** Aug 2018 – June 2020
M.A. in Physics
- **Indian Institute of Technology Roorkee** July 2014 – May 2018
B.Tech., Electronics and Communication Engineering, Minor in Physics

RESEARCH EXPERIENCE

- **Graduate Research Assistant, C.N. Yang Institute for Theoretical Physics** Aug 2020 – Present
Stony Brook University
 - Constructed dimension-eight operator basis for universal effective field theories, cross-checked using group representation theory techniques to compute group invariants.
 - Designed and implemented Monte Carlo simulation workflows generating **1M+** events for collider physics.
 - Performed symbolic manipulations using Mathematica and multi-parameter regression via log-likelihood minimization using simplex algorithm in Python and C++.
 - Collaborated in multi-institutional research teams; co-authored multiple peer-reviewed publications in Physical Review D journal.
- **Research Assistant, Center for Nuclear Science** Aug 2018 – June 2020
Stony Brook University
 - Experience with large-scale simulation packages generating **1M+** events and C++ based data analysis framework ROOT, executed via HTCondor job submission on the Brookhaven National Lab computing cluster.
 - Applied statistical reweighting techniques to estimate uncertainty reduction projections for the proposed electron-ion collider.

PUBLICATIONS

All papers can be found on my [INSPIRE](#) and [Google Scholar](#) profile.

- “Drell-Yan production in universal theories beyond dimension-six SMEFT”
[arXiv:2503.19962 \[hep-ph\]](#), *Phys. Rev. D* **112**, 013009 (2025)
Tyler Corbett, **Jay Desai**, O. J. P. Eboli, M. C. Gonzalez-Garcia, Matheus Martinez, Peter Reimitz
- “Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
[arXiv:2404.03720 \[hep-ph\]](#), *Phys. Rev. D* **110** (2024) 033003
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia
- “Impact of dimension-eight SMEFT operators in the EWPO and Triple Gauge Couplings analysis in Universal SMEFT”
[arXiv:2304.03305 \[hep-ph\]](#), *Phys. Rev. D* **107**, 115013 (2023)
Tyler Corbett, **Jay Desai**, O.J.P. Eboli, M.C. Gonzalez-Garcia, Matheus Martinez, Peter Reimitz

RELEVANT PROJECTS

- **Quantum Programming Project — Quantum Error Correction**
Stony Brook University
 - Implemented and analyzed quantum circuits in IBM Qiskit to study quantum error correcting codes and fault-tolerant computation in a graduate-level quantum programming course.
 - Investigated the Nishimori phase transition in quantum error correcting codes, re-implementing IBM Quantum simulations based on recent research ([arXiv:2309.02863](#)) to analyze error thresholds.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java
- **Scientific Computing:** NumPy, Mathematica, MATLAB, ROOT, IBM Qiskit
- **Machine Learning & Data Science:** Matplotlib, Scikit-learn, Pandas, Jupyter, statistical inference, multi-parameter optimization, high-dimensional modeling
- **High-Performance Computing:** HTCondor, Linux cluster computing
- **Simulation & Modeling:** Monte Carlo Methods: MadGraph
- **Tools:** Linux, Version Control, LaTeX

SEMINARS/CONFERENCE TALKS

- **Brookhaven National Lab (BNL) Seminar (invited), Brookhaven, December 11th 2025**
“Universal Standard Model Effective Field Theory beyond dimension-six”
- **Higgs and Effective Field Theory (HEFT), CERN, June 3rd 2025**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **LHC EFT WG meeting, (online), April 28th 2025**
“Drell-Yan production in universal theories beyond dimension-six SMEFT”
- **DPF-Pheno 2024, University of Pittsburgh, May 13th 2024**
“Dimension-eight Operator Basis for Universal Standard Model Effective Field Theory”
- **Pheno 2023, University of Pittsburgh, May 8th 2023**
“A Phenomenological study of Higgs jets at a muon collider”
- **Higgs 2022, Pisa, Italy, Nov 9th 2022(hybrid)**
“A Phenomenological study of Higgs jets at a muon collider”
- **EIC Yellow Report: semi-inclusive working group meeting**
“Impact study of Kaon SIDIS at EIC on polarized strange quark distribution function”
- **APS April Meeting, 2020**
“Strange quark polarized distribution function using SIDIS at EIC”

AWARDS & ACHIEVEMENTS

- Gerald Brown Prize (Research Travel Award), Stony Brook University
- Nationwide Education and Scholarship Test (Ranked **13th** nationally)
- Joint Entrance Examination (JEE) Advanced (Ranked **971** nationally)
- Selected for **Indian National Astronomy Olympiad (INAO-2014)**; top 1% in **National Standard Examination in Astronomy - 2014**.
- Selected for **Regional Maths Olympiad (RMO) - 2014**.

LEADERSHIP & TEACHING

- Instructor and Teaching Assistant for undergraduate physics courses.
- Journal Referee for the Journal of High Energy Physics.